

Human Computer Interaction**CSE 407 Term Exam Fall-2024**

October 10, 2024

Max Marks: 20

Duration: 2 hours

Notes:

1. Attempt all questions. The paper comprises 6 questions.
2. You are suggested to try to attempt the questions in the given sequential order of the paper.
3. Be succinct

- Q1) Consider the following screen given in figure 1, only computer scientists or experts can work through this screen, what is the basic reason that why all humans cannot work through this screen? if a user is working through this screen, then his/her working memory load is high as user must recall all commands required for that task. If you are asked to make a slight change to this screen where the user working memory load can be reduced, how will you change it? Which design principle will help you? Sometimes we move our data from one drive to other by writing a command and then press enter, after pressing the enter we are not sure that whether our task is in progress or not? How this problem can be resolved through a design principle of learnability. (4 marks).

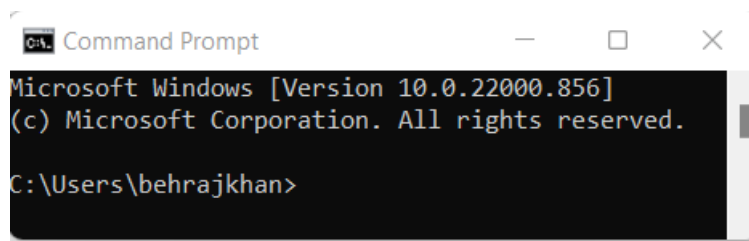


Figure 1: Dialogue Box for Q1

- Q2) As we have discussed about the design principles along with examples. In last week I asked a question from two students while they were talking to each other, but both were unable to even recall my question. Write down the reason behind this phenomenon that why both were unable to just repeat my question. Let us consider that you want to design a application to avoid this kind of problems in near future for your friend. You decided to warn your friend through his/her cellphone through your designed app. Which echoic channels of your friend will be helpful in receiving the warning. You tried but your friend was unable to notice the warning and keep arguing with you about not informing him/her after the class. Which implemented design principle will be help you out as proof that you have tried to inform him/her, but he/she did notice? As some of the students were arguing about the solution of the mentioned scenario as one of them were absent on that day that's why she was unable to recall as information weren't at long term memory. Consider the absence scenario and explain that why she was unable to find the solution in her long-term memory.

Note (writing Absence is not applicable, you must answer from memory perspective that how and what are necessary for moving info to long term memory). (4 marks)

- Q3) One of the Shneiderman's rule "Design dialogue to yield closure", is applied in the following figure 3, it can be implemented by using one of flexibility principle. Mention that principle and also explain the reason for applying this concept in particular figure. (2 marks)

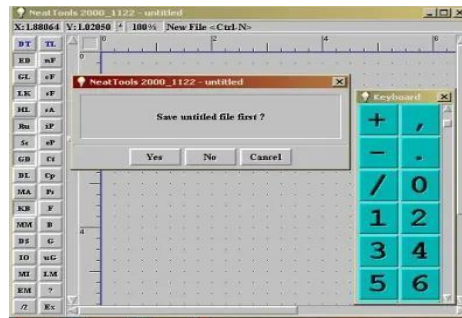


Figure 2: Dialogue box for Q3

- Q4) Consider the screen given below of daraz.pk and list down metaphors used in the screen (figure 4). Why are metaphors used? How it helps the user. If metaphors are missing, then what kind of problems can occur? Also mention that whether multithreading or multitasking is implemented in this screen? Is multimodality applied? Name the function where immediate honesty is applied? Rationalize. (4 marks)

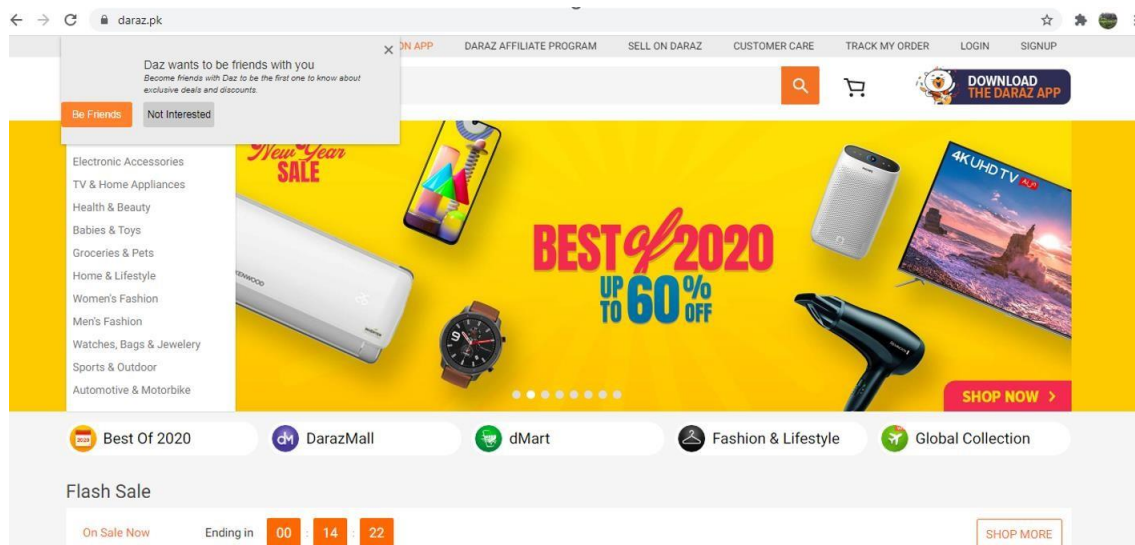


Figure 4: Dialogue box for Q4

Q5)

Fitt's law is used to measure the time required for a person to move a pointer to a selected option. Microsoft believes that the Ribbon Menu shown in figure 5 takes less time as compared to a drop-down menu, shown in figure 5. Do you agree with this statement or not? Justify your answer by giving proper arguments. (4 marks)

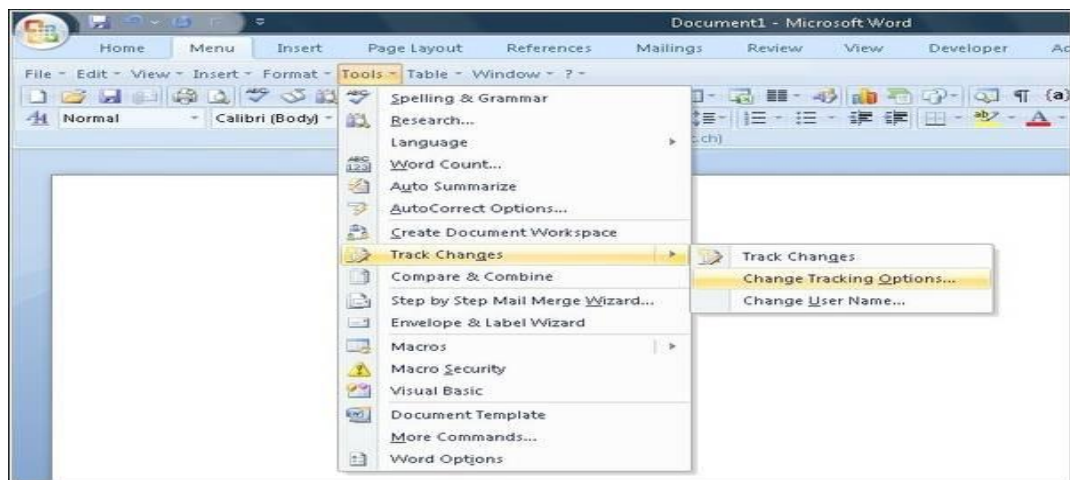


Figure 5: Dialogue box for Q5

Q6)

Mention a window screen of your HCI semester project where usability principle “task conformance” is missing. Redesign the mentioned screen by implementing the mentioned principle. Also assume that you are buying an item online through your chosen financial app. You have noticed one of the common interface mistakes “inflexible input” while entering the credit card detail. How will you overcome this problem? (2 marks)